



ECE748 Final Project

Differential common gate transimpedance bandpass filters (TI-BPFs)

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Objectives:

The objective of this design is to develop a transimpedance bandpass filter based on a differential common gate amplifier by following symbolic synthesis methodologies illustrated in [1]. This approach will utilize two-port network representations and symbolic mathematical tools to ensure accurate consideration of nonidealities and parasitics, ultimately leading to efficient and high-performance filter solutions.

Problem Statement:

Designing high-performance analog circuits, such as transimpedance (TI) bandpass filters, often involves complex and time-consuming manual processes. Traditional approaches may overlook optimal circuit configurations and struggle to account for nonidealities and parasitics, leading to suboptimal performance in real-world applications. In particular, TI bandpass filters are essential in systems requiring current-to-voltage conversion and selective frequency amplification, including optical communication, biomedical sensing, and spectroscopy. However, achieving accurate control over gain, bandwidth, and noise in these circuits is challenging.

To address this, a systematic synthesis methodology using symbolic math tools can simplify the design process by automating circuit generation and analysis. This project seeks to apply such a methodology to create TI bandpass filters based on a differential common gate amplifier topology. The goal is to explore innovative circuit configurations and optimize performance for practical applications, ensuring robust, and efficient designs with minimal manual intervention.

Introduction:

Analog circuit design plays a critical role in various electronic systems, requiring accurate control over performance parameters such as gain, bandwidth, and noise. The synthesis methodology adopted in this project enables the systematic and automated generation of analog circuits using symbolic tools. By following the two-port network matrix representation of MOS transistors, this approach facilitates the exploration of various circuit configurations while accounting for nonidealities and parasitics, significantly reducing manual design effort [1].

This project focuses on designing transimpedance bandpass filters (TI-BPF) based on a differential common gate amplifier topology. TI-BPFs are critical in applications where signals need to be converted from current to voltage while selectively amplifying a desired frequency band. TI-BPF can be used in various domains, including:

- Optical communication systems, where they process photodiode currents in optical receivers.
- Biomedical instrumentation, for amplifying small bio-signals, such as those from photodetectors in pulse oximeters and other medical sensors.

- Spectroscopy and sensing applications, where they enhance signal quality by filtering out unwanted frequencies.
- Wireless communication, to improve selectivity and reduce interference in front-end circuits.

By adopting a symbolic synthesis approach, the project aims to develop high-performance TI-BPFs optimized for practical applications.

This report presents design examples along with their corresponding simulations. First, the general analysis of a transimpedance bandpass filter is discussed. Subsequently, two design examples are provided, accompanied by their simulation results for illustration.

Analysis:

A transimpedance bandpass filter (TI-BPF) converts an input current signal into a voltage output while allowing only a specific frequency band to pass through, effectively rejecting signals outside the desired range. The transfer function $H(s)$ of a TI-BPF typically takes the form:

$$H(S) = \frac{V_o}{I_{in}} = \frac{K \cdot s}{s^2 + \frac{w_o}{Q}s + w_o^2}$$

Where, where K is a gain term, w_o is the center angular frequency, and Q is the quality factor. Poles and zeros of the transfer function play a critical role in shaping the frequency response. Zeros determine the frequencies at which the output is attenuated, while poles define the resonant frequencies and bandwidth. The quality factor Q is representing the selectivity of the filter, and it is given by:

$$Q = \frac{w_o}{BW}, \text{ hence } Q > 1, \text{ narrow band}, Q < 1 \text{ wide band}$$

As shown in Fig. 1, six impedances can be used to build a common gate differential filter, and based on the nature of each impedance (i.e., capacitor, inductor, or resistor) a different filter can be generated. From [1], a transmission matrix (i.e., ABCD matrix) is used to model each transistor, and using symbolic math toolbox, the general transfer function for the TI-BPF is generated as follows:

$$H(s) = \frac{V_o^+ - V_o^-}{I_{in}^+ - I_{in}^-} = \frac{N(s)}{D(s)}$$

$$N(s) = (Z_5(a_{11} - 1) + Z_5a_{12} - Z_2a_{12})$$

$$\begin{aligned}
D(S) = & Z_2 Z_5 (a_{21} + a_{11} a_{22} + a_{11} + a_{12} a_{21} - a_{22} - 1) \\
& + 2Z_2 (a_{11} a_{22} + 2a_{11} + a_{12} a_{21} - a_{22} - 2) \\
& + \frac{Z_2}{Z_L} [a_{12} + Z_5 (a_{11} a_{22} + a_{11} + a_{12} a_{21} - a_{22} - 1)] + \frac{1}{Z_L} [a_{12} + Z_5 a_{12} + 4a_{12}] \\
& + \frac{Z_2}{Z_3 Z_L} [a_{12} + Z_5 a_{12}] + \frac{Z_2 Z_5}{Z_L Z_4} [a_{11} a_{22} + a_{11} + a_{12} a_{21} - a_{22} - 1] \\
& + Z_5 (a_{12} + 4a_{12}) + 2Z_2 Z_5 (a_{11} a_{12} + a_{12} a_{21} - a_{22} a_{11}) \\
& + \frac{2Z_2 Z_5}{Z_4} (a_{11} a_{12} + a_{22} a_{21}) + \frac{Z_5}{Z_L} (a_{12} a_{21} - a_{22} a_{11})
\end{aligned}$$

Where a_{ij} are the coefficients of the transmission matrix ABCD, which are dependant on the level of nonidealities considerations as follows:

1. Complex transmission matrix:

$$\begin{bmatrix} \left(\left(\frac{1}{r_o} + s * C_{gd} \right) \right) & \left(\frac{1}{s * C_{gd} - gm} \right) \\ \left(\left(\frac{((s * C_{gd} C_{gs} r_o) + (C_{gd} gm r_o) + C_{gd} + C_{gs}) * s}{s * C_{gd} - gm} \right) \right) & \left(\frac{s * (C_{gs} + C_{gd})}{s * C_{gd} - gm} \right) \end{bmatrix}, \text{Considering: } \lambda, C_{gs}, \text{ and } C_{gd}.$$

2. Simple transmission matrix: $\begin{bmatrix} 0 & \frac{-1}{gm} \\ 0 & 0 \end{bmatrix}$, ignoring all non ideal effects

For example, if Z_L is a parallel combination of an inductor and a capacitor, and Z_3 and Z_1 are resistors, the filter transfer function is as follows:

$$H(s) = \frac{V_o^+ - V_o^-}{I_{in}^+ - I_{in}^-} = \frac{L_L R_1 R_3 g m s}{(R_1 g m + 1)(C_L L_L R_3 s^2 + L_L s + R_3)}, \text{ then } w_o = \frac{1}{\sqrt{L_L C_L}}, Q = \frac{C_L R_3}{\sqrt{L_L C_L}}, \text{ considering the simple transmission matrix.}$$

Considering the combinations of two and three impedances, a total of 407 TI-BPF configurations can be generated. The number of possible TI-BPFs for each combination is detailed in Table 1.

To optimize area usage, topologies with a single inductor and a single capacitor in each branch (i.e., a maximum of two inductors and two capacitors in the entire circuit) were selected for design and fabrication in future work. This approach minimizes area consumption, as inductors and capacitors are among the most space-demanding components.

Table 1 The number of possible TI-BPFs for each impedances' combination

Number of possible TI-BPF	Combinations of impedances
22	$Z_1, Z_3,$ and Z_L
6	$Z_1, Z_2,$ and Z_L
22	$Z_1, Z_4,$ and Z_L
6	$Z_1, Z_5,$ and Z_L
4	$Z_1,$ and Z_L
104	$Z_2, Z_3,$ and Z_L
104	$Z_2, Z_4,$ and Z_L
2	$Z_2, Z_5,$ and Z_L
8	$Z_2,$ and Z_L
91	$Z_3, Z_4,$ and Z_L
16	$Z_3, Z_5,$ and Z_L
13	$Z_3,$ and Z_L
16	$Z_4, Z_5,$ and Z_L
13	$Z_4,$ and Z_L
2	$Z_5,$ and Z_L
407	Total TI-BPF

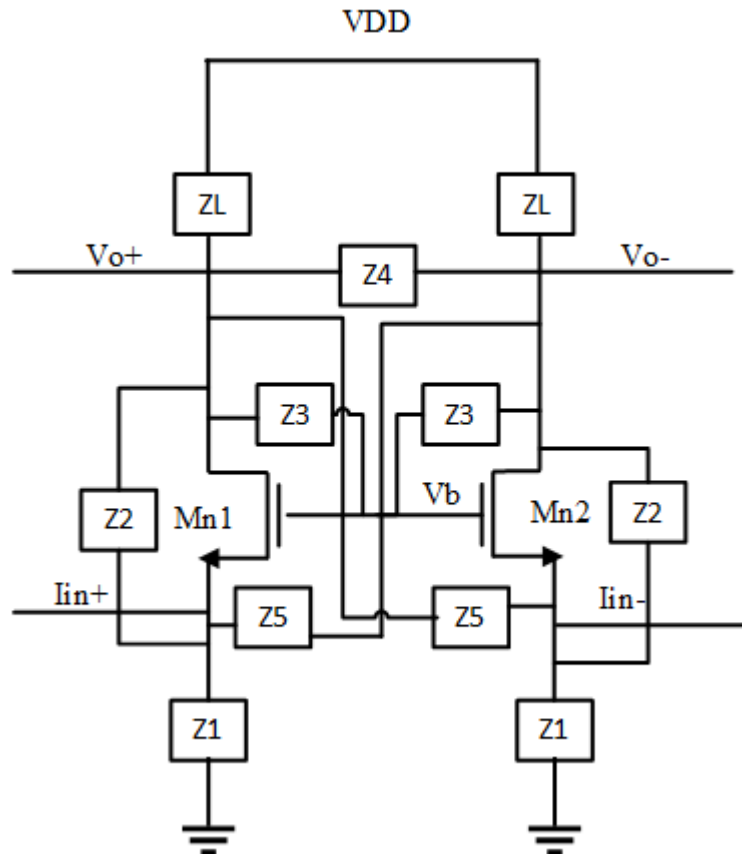


Fig. 1 Common gate TI-BPF with all possible impedance combinations

Design Examples:

1. A combination of $Z_{1,4}$ as resistors, and Z_L as a parallel combination of a capacitor and inductor

As shown in Fig. 2, a TI-BPF-1 can be generated using the three impedances $Z_{1,4,L}$ given that $Z_{1,4}$ are resistors and Z_L is a parallel combination of an inductor and a capacitor. The transfer function of this filter (i.e., using the simple transmission matrix) is as follows,

$$H(s) = \frac{L_L R_4 R_1 g_{ms}}{(R_1 g_m + 1)(C_L L_L R_4 s^2 + 2L_L s + R_4)}, w_o = \frac{1}{\sqrt{L_L C_L}}, Q = \frac{C_L R_4}{2} \frac{1}{\sqrt{L_L C_L}}$$

To design for a central frequency of 10 GHz, let's assume that C_L is 400 fF, and hence L_L is 633.3 pH. R_4 can be tuned to change the filter selectivity, either wide band or narrow band and to increase the gain of the filter. Transistors have aspect ratio of 1 μ m/60nm, the supply voltage (V_{DD}) and the gate bias for the transistors (V_b) are 1 V, and R_1 is 600 Ω . The current consumed in each branch is about 249 μ A, hence the total power consumption is almost 0.5 mW.

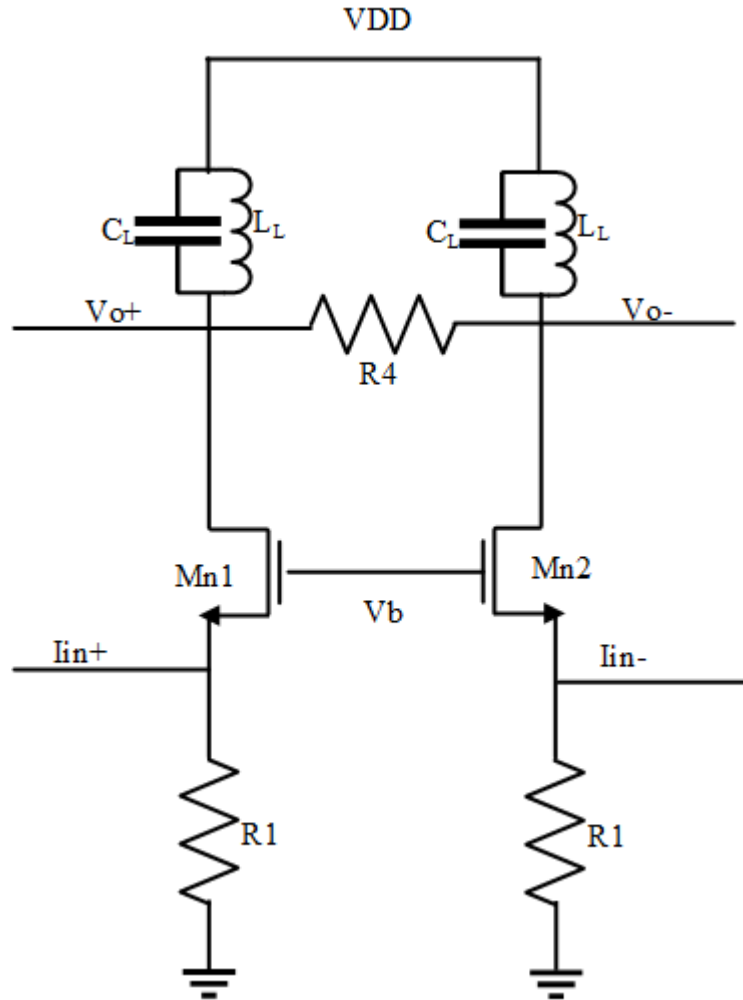


Fig. 2 TI-BPF-1 a combination of parallel capacitor and inductor in the load, R_1 , and R_4

2. A combination of Z_1 as an inductor, and Z_L as a parallel combination of capacitor and resistor

As shown in Fig. 3, a TI-BPF-2 can be generated using the three impedances $Z_{1,L}$ given that Z_1 is inductor and Z_L is a parallel combination of a resistor and a capacitor. The transfer function of this filter (i.e., using the simple transmission matrix) is as follows,

$$H(s) = \frac{L_1 R_L g_m s}{(C_L R_L s + 1)(L_1 g_m s + 1)}, w_o = \frac{1}{\sqrt{L_1 C_L R_L g_m}}, Q = \frac{L_1 C_L R_L g_m}{\sqrt{L_1 C_L R_L g_m}}$$

The difference between this filter and the first one lies in its tunability. By adjusting the values of R_L , the center frequency can be modified, making this filter adaptable for multiple applications. This tunability provides flexibility and expands the range of applications where the filter can be effectively employed.

To achieve tunability, R_L can be designed as a PMOS transistor with a gate voltage V_R , then when changing the gate voltage, the value of R_L is changed.

If C_L is 500 fF and L_1 is 200 nH, the center frequency (w_o) will depend on the value of the load resistance (R_L) and the transconductance (g_{mN}) of the NMOS transistor. The NMOS transistors have an aspect ratio of $0.8\mu\text{m}/60\text{nm}$, while the PMOS transistors have an aspect ratio of $19.5\mu\text{m}/60\text{nm}$. The supply voltage (V_{DD}) and the gate bias for the transistors (V_b) are set to 1 V. The gate voltage of the PMOS transistors can vary between 0 V and 590 mV, allowing the value of the active load to be tunable while keeping NMOS in saturation region.

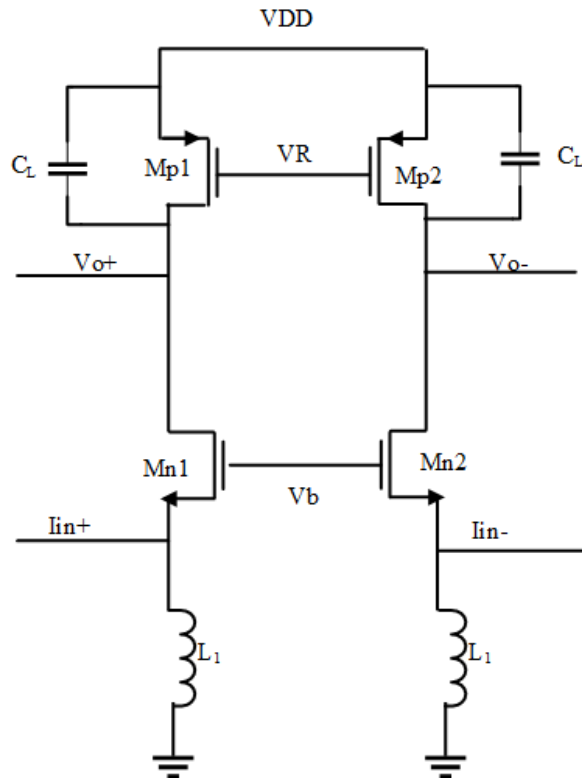


Fig. 3 TI-BPF-2 a combination of L_1 , and parallel combination of C_L and R_L

Pre- layout simulations:

1. TI-BPF-1 A combination of $Z_{1,4}$ as resistors, and Z_L as a parallel combination of a capacitor and inductor

In this section, pre-layout simulations for the TI-BPF-1 are presented. In these simulations, $R_{1,4}$ is modeled with `rpoly_rf`, C_L is modeled using `MIMcap`, and L_L is modeled using `spiral_sym_mza_a` from the design kit. The inductor has a quality factor of 13 at frequency of 10 GHz.

As illustrated in Fig. 4, the center frequency f_0 is approximately 10 GHz. Adjusting R_4 does not affect the center frequency; however, it significantly impacts the filter's selectivity and gain, as predicted by the filter's transfer function.

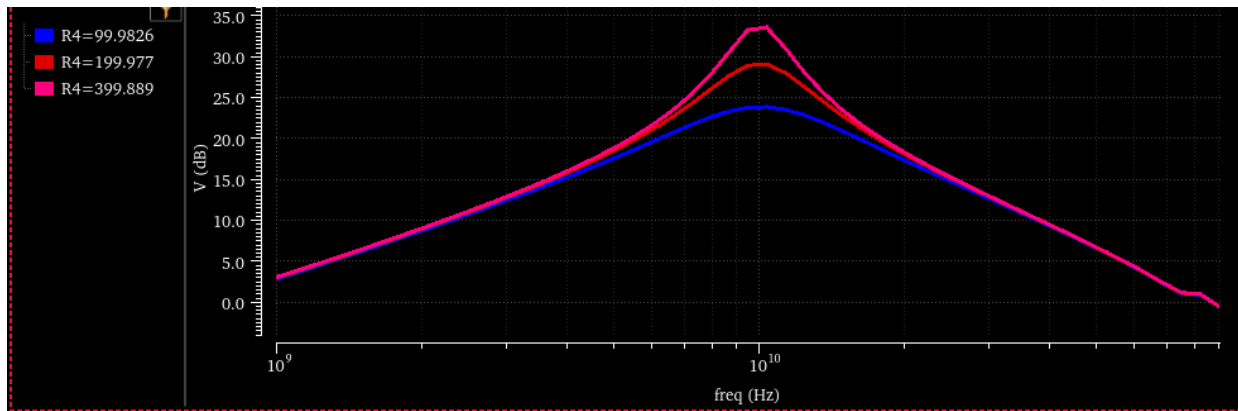


Fig. 4 TI-BPF-1 AC response with different values of R_4

As shown in Fig. 5, the output voltage signal of the TI_BPF-1 is completely attenuated when the frequency of the input signal is 100 MHz, and partially attenuated when the frequency of the input signal is 5 GHz; however, the output signal exists, when the frequency of the input signal is 10 GHz.

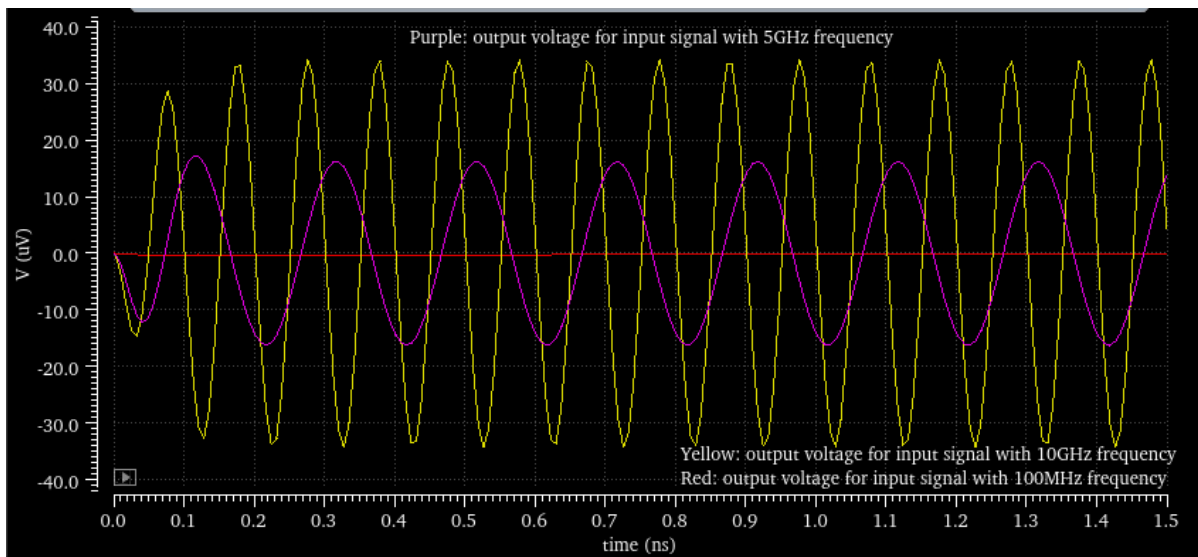


Fig. 5 TI-BPF-1 output voltage with different frequencies for the input signal

When considering PVT variations, it is found that the AC response is changed slightly with temperature and voltage variations (see Fig. 6), as the peak gain changed from 27 dB to 29.8 dB when change temperature from -40°C to 125°C and change the voltage from 0.9 V to 1.1 V, given that the gain at room temperature and supply voltage of 1 V is 29 dB. However, the center frequency and the BW of the filter are almost the same.

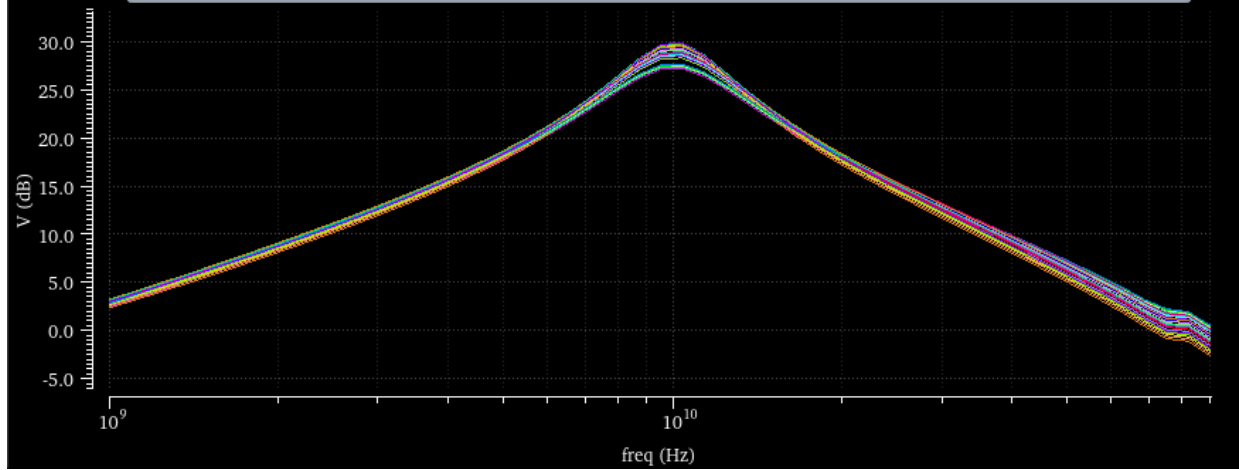


Fig. 6 TI-BPF-1 AC response in the presence of voltage and temperature variations

2. TI-BPF-2 A combination of Z_1 as inductor, and Z_L as a parallel combination of a capacitor and an active load

In this section, the pre-layout simulations for the TI-BPF-2 are presented. During these simulations, the load capacitance (C_L) is modeled using a MIM capacitor (MIMcap), while the load inductance (L_L) is represented as an ideal inductor. Due to its large value, L_L is assumed to be implemented as an off-chip inductor (can not find an inductor in the tsmc library that support this large value).

As shown in Fig. 7, the central frequency of the TI-BPF-2 shifts with changes in the value of V_R . Specifically, as V_R varies from 0.2 V to 0.512 V, the central frequency (f_0) changes from 1.6 GHz to 501 MHz. Additionally, the DC gain of the circuit is affected due to its dependency on the load resistance.

It should be stated that, the bias voltages V_b and V_R are implemented using diode connected transistors, and ideal current sources.

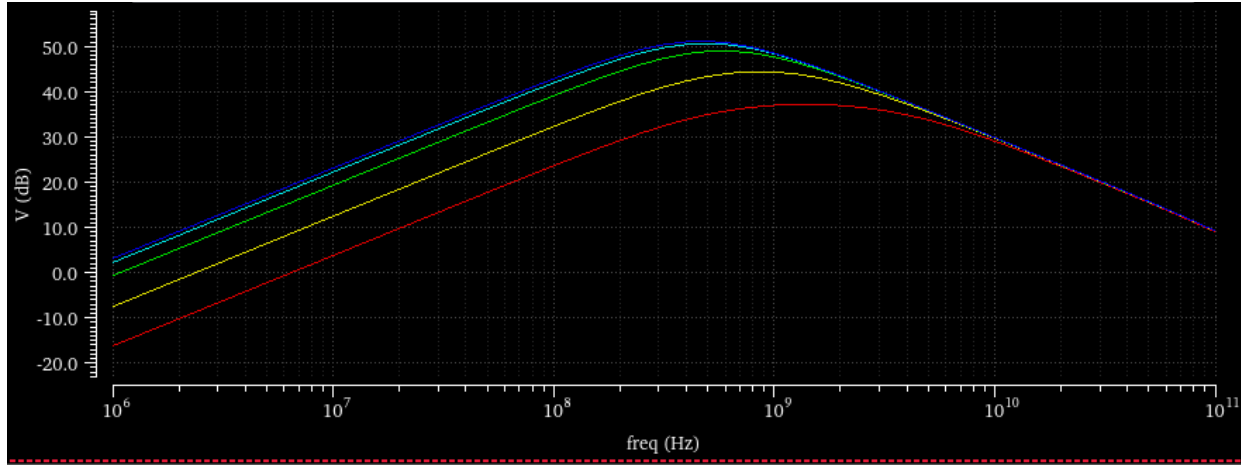


Fig. 7 TI-BPF-2 frequency response with different VR (from 200 mV to 512 mV)

As shown in Fig. 8, when considering V_R equals to 0.5 V, where f_o is about 501 MHz, the output voltage signal of the TI_BPF-2 is attenuated when the frequency of the input signal is 10 GHz, and 100 MHz; however, the output signal exists, when the frequency of the input signal is 500 MHz.

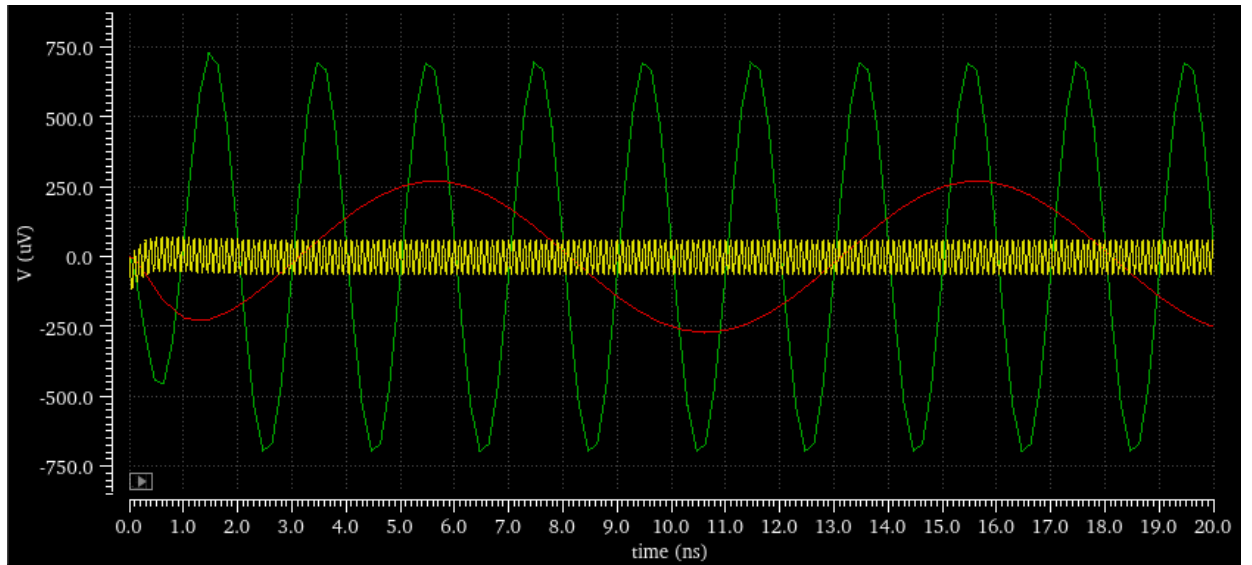


Fig. 8 TI-BPF-2 output voltage with different frequencies for the input signal

When considering process, voltage, and temperature (PVT) variations (i.e., temperature from -40°C to 125°C and $\pm 10\%$ variations in voltages), it is observed that the AC response of the circuit is significantly affected by changes in temperature and biasing voltages, as shown in Fig. 9.

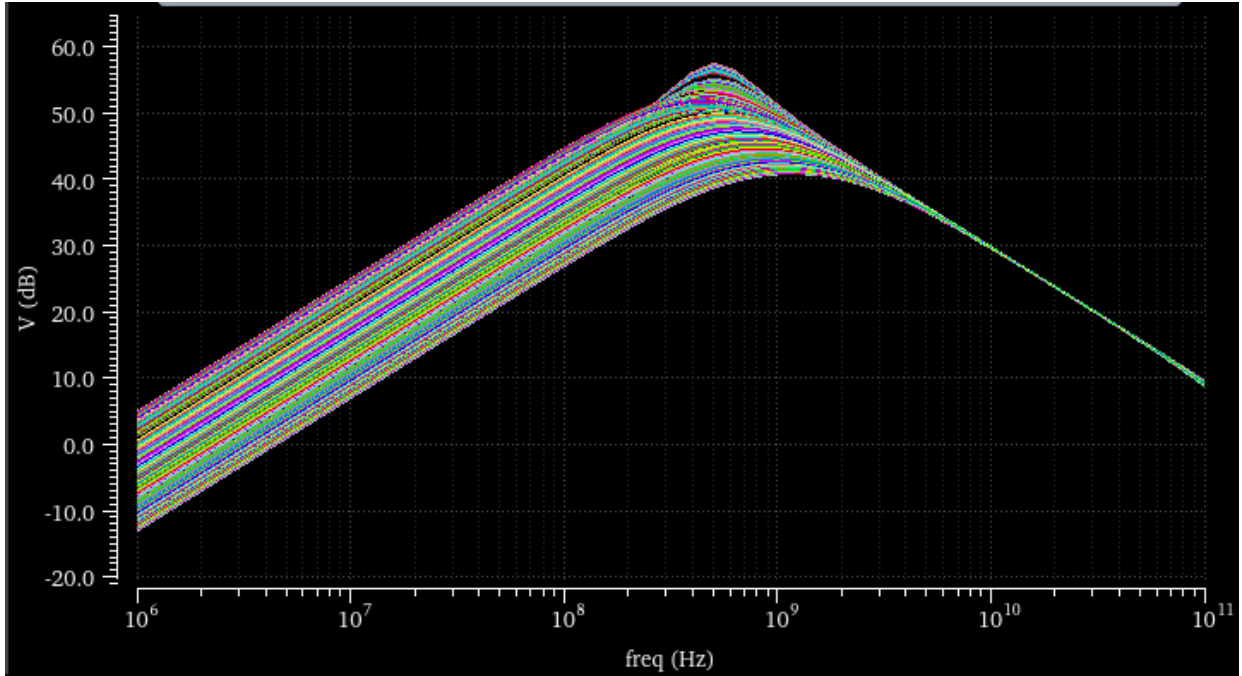


Fig. 9 TI-BPF-2 AC response in the presence of voltage and temperature variations

Under nominal conditions ($V_R=0.5$ V, $V_{DD}=1$ V, and $V_b=1$ V), the central frequency is 501 MHz, and the DC gain is 51 dB. However, with temperature varying from -40°C to 125°C , supply voltage fluctuating by $\pm 10\%$, and the DC currents that used in biasing the gates fluctuating by $\pm 10\%$, the transistors the central frequency and DC gain exhibit significant changes:

- **DC Gain:** Varies from 40.8 dB to 57.5 dB.
- **Central Frequency:** Shifts from 501 MHz to 1.26 GHz.

These significant variations are primarily caused by:

1. **Load Resistance Sensitivity:** Variations in V_R alter the load resistance, impacting both the central frequency and the gain.
2. **Transconductance (g_m) Sensitivity:** Changes in V_b affect the transconductance of the transistor, further influencing the AC response.

The tunability of the design, while beneficial for adapting to different operating conditions, makes the circuit highly sensitive to voltage variations. This sensitivity highlights the importance of careful design optimization and calibration to minimize the impact of PVT variations on circuit performance.

Under nominal conditions ($V_R=0.5$ V, $V_{DD}=1$ V, and $V_b=1$ V), the circuit consumes total current of 1.2 mA, which result on a total power consumption of 1.2 mW.

Post-layout simulations:

TI-BPF-2 A combination of Z_1 as inductor, and Z_L as a parallel combination of a capacitor and ac active load

In this section, the post-layout simulations of the proposed design are presented. As the inductor has a large value, it is assumed to be off chip. As shown in Fig. 10, the chip area measures approximately $55 \mu\text{m} \times 47 \mu\text{m}$, with the majority of the area occupied by the capacitor (500 fF). An AC response simulation was performed to evaluate the impact of parasitic effects on the filter's performance. As shown in Fig. 11, under same biasing conditions ($V_{DD}=V_b=1 \text{ V}$, and $V_R=0.5 \text{ V}$) the center frequency shifts from 501 MHz to 794 MHz, while the DC gain decreases from 51 dB to 43.5 dB. Additionally, due to the parasitic effect, a new zero is introduced in the transfer function, resulting in a constant gain starting from 20 GHz to 90 GHz. This behaviour is predicted as the numerator of the transfer function of this configuration when considering all parasitics is $N(s) = Z_1 Z_L (C_{gd}s - gm)(gmro + 1)$ and here Z_1 and Z_L depend on frequency, which introduces more zeros in the transfer function. It is important to note that this zero lies outside the filter's 3-dB bandwidth, as it starts with a gain of 30 dB, while the filter's DC gain is 43.5 dB.

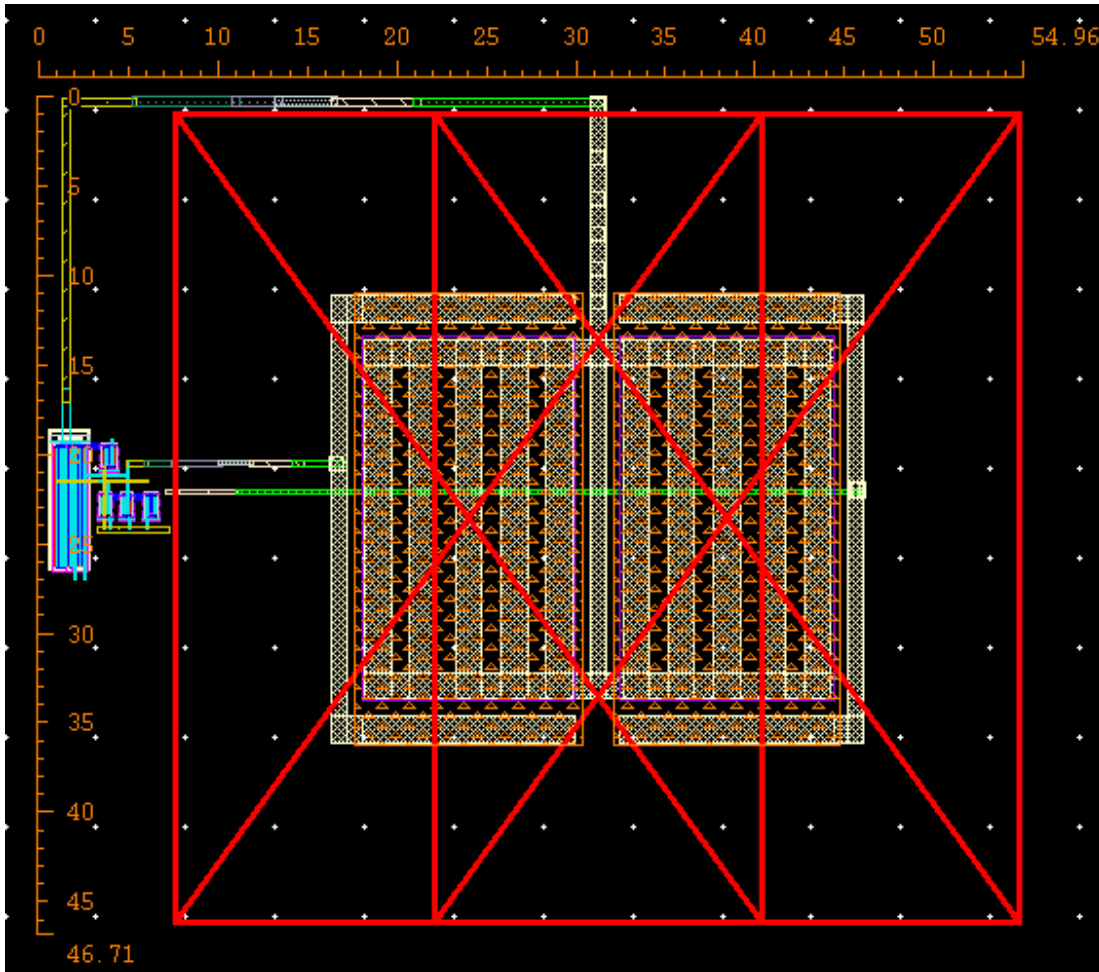


Fig. 10 TI-BPF-2 Layout

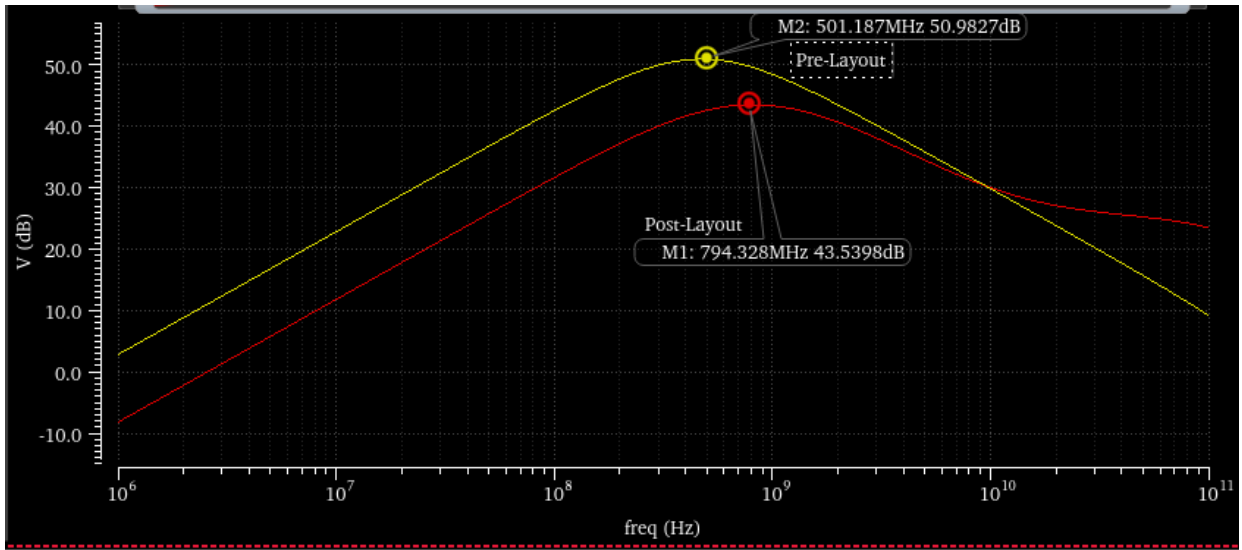


Fig. 11 TI-BPF-2 pre-layout and pos- layout frequency response with different V_R is 500 mV and V_G is 1 V. In order shift the center frequency back to 501 MHz (see Fig. 12), V_R should be set to 0.565 V. This highlights another advantage of tunability, as it allows compensation for variations caused by parasitic effects or PVT changes.

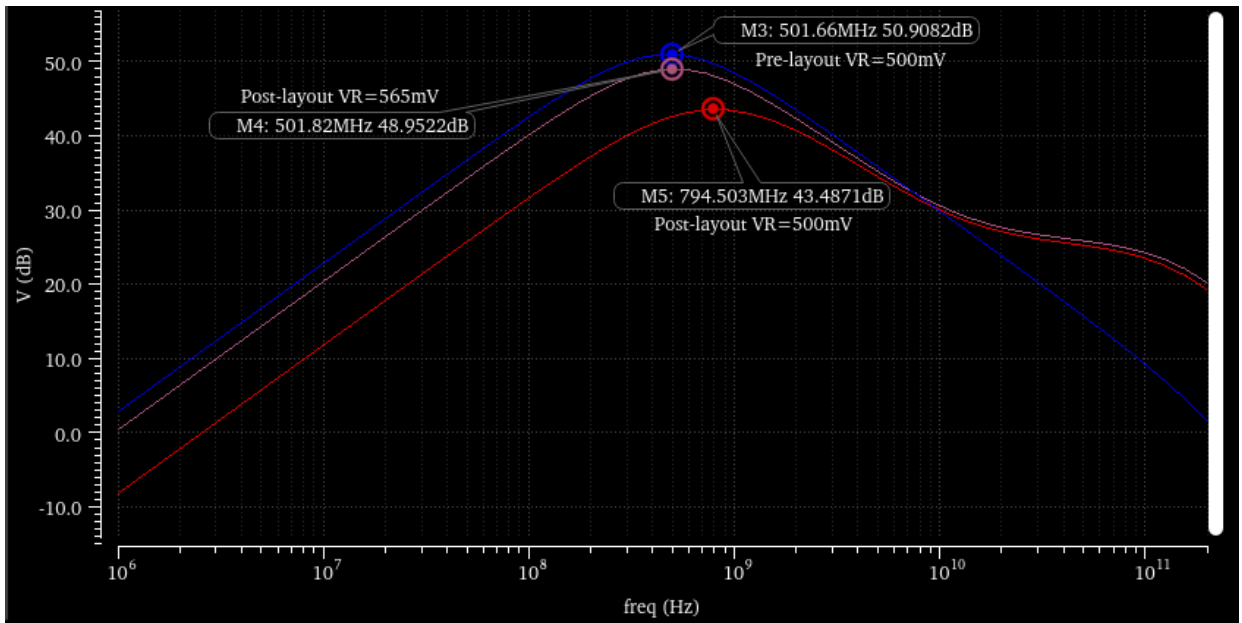


Fig. 12 Changing V_R to achieve same W_o in post-layout simulations for TI-PBF-2. With variations in temperature from -40°C to 125°C , fluctuations in supply voltage by $\pm 10\%$, and the DC currents that used in biasing the gates by $\pm 10\%$, the central frequency and DC gain exhibit significant changes (see Fig. 13):

- **DC Gain:** Varies from 33 dB to 53.8 dB.
- **Central Frequency:** Shifts from 450 MHz to 1.5 GHz.

Not that, with the advantage of tunability, any variations caused by temperature or voltage fluctuations can be effectively compensated.

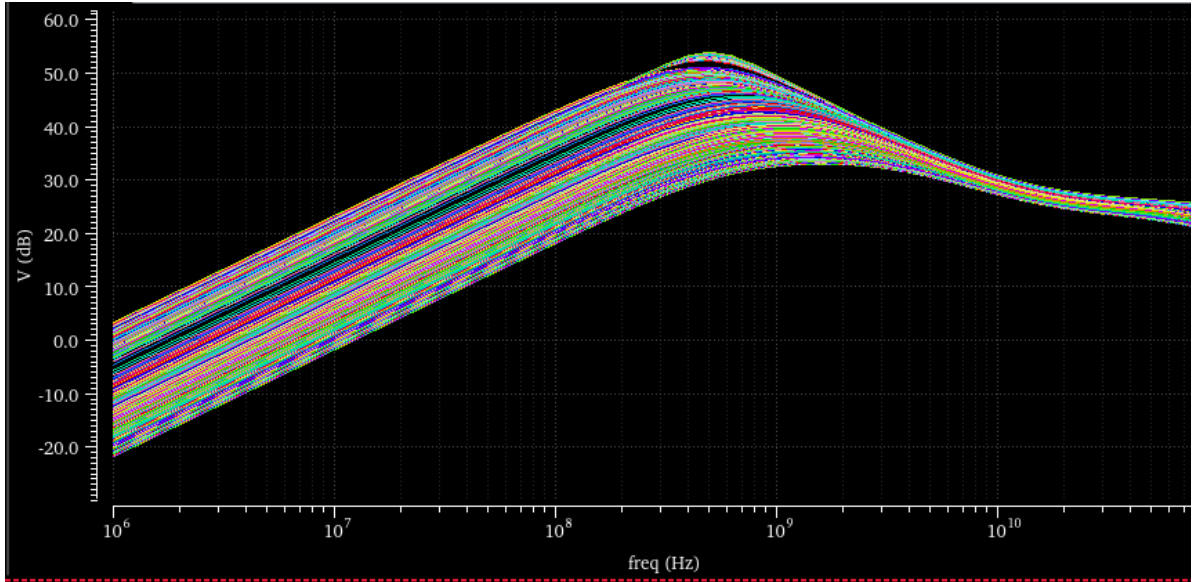


Fig. 13 TI-BPF-2 AC post-layout response in the presence of voltage and temperature variations

Comparison

In [2], a transimpedance MOS-C bandpass filter is presented, employing the transconductance (g_m) of MOS transistors and capacitors to achieve a bandpass filter behaviour. The center frequency is determined by the transistor g_m and the capacitor values, and tunability can be achieved by adjusting the transistor biasing; however, this capability for tunability is not mentioned in the paper. Simulations using LT-spice with a TSMC 180 nm model show a center frequency of 1.13 MHz and a DC gain of 90 dB, with a power consumption of 7.8 μ W. The design uses two capacitors, 1 pF and 10 pF to achieve the above-mentioned center frequency.

The proposed TI-BPF-2 uses capacitors, inductors, and active loads in a differential common-gate topology to realize bandpass filter functionality. With an inductor of 200 nH and a capacitor of 400 fF, the design achieves a center frequency of almost 500 MHz. This frequency can be tuned by adjusting the gate bias of the active load (modifying R_L) and the main transistor (modifying g_m), providing two tunability factors. Pre-layout and post-layout simulations are performed using CMOS 65 nm technology. The power consumption of approximately 1.2 mW when V_R is 0.5 V, V_b , and V_{DD} are 1 V. The differential pair topology of the proposed filter offers significant advantages, including improved noise immunity and a larger output swing.

Discussion and conclusion

A total of 407 TI-BPF designs can be generated using combinations of only two or three impedances from a set of six possible impedances. However, not all these filters possess tunability,

as the center frequency in some designs depends solely on the values of inductors and capacitors. Additionally, some configurations require more than two inductors or capacitors in the differential topology, which increases complexity and area requirements.

The two examples discussed in this report are optimized for minimal chip area, utilizing only two inductors and two capacitors (one per branch). In the first design, the center frequency is inversely proportional to the inductor and capacitor values, making it suitable for high-speed data communication applications where smaller inductor and capacitor values are required (i.e., in pH and fF).

In contrast, the second configuration achieves a tunable center frequency that depends not only on the values of L_1 and C_L but also on the transconductance (g_m) of the common-gate transistor and the load resistance. The load resistor is implemented as an active load with external gate biasing, allowing its value to be adjusted for the desired center frequency and providing tunability to accommodate PVT variations. Furthermore, the transistor's g_m can also be tuned through its gate bias, enhancing the design's flexibility.

The proposed TI-BPF configuration is highly flexible and well-suited for optical data communication applications. It can function as a transimpedance amplifier (TIA) and a filter, or as a combined TIA and continuous-time linear equalizer (CTLE) within a single circuit. By introducing a non-zero zero to the design, the TI-BPF can perform the CTLE function to compensate for channel losses while simultaneously amplifying the received signal. This dual functionality allows the TI-BPF to replace two separate circuits in the receiver chip, simplifying the design and improving efficiency (see Appendix).

Moreover, the TI-BPF is advantageous in terms of area efficiency, particularly in high-speed applications where the center frequency is high (e.g., in the 10s of GHz). In such scenarios, the reduced inductor and capacitor values required for higher frequencies result in smaller overall circuit size.

This makes the TI-BPF an excellent choice for space-constrained receiver-side implementations in optical communication and high-speed data communication systems.

Future work

Here are some points that can be considered for future work:

1. Investigation of TI-BPF Conversion to Function as a CTLE with a TIA:

By modeling an optical channel with specific loss characteristics and design the TI-BPF to include a non-zero "zero" at the same frequency as the channel's pole. This would enable the TI-BPF to function effectively as both a transimpedance amplifier (TIA) and a continuous-time linear equalizer (CTLE). As demonstrated in Appendix Fig. 2, the gain provided by such a TI-BPF functioning as a TIA with CTLE is sufficient for high-speed optical communications.

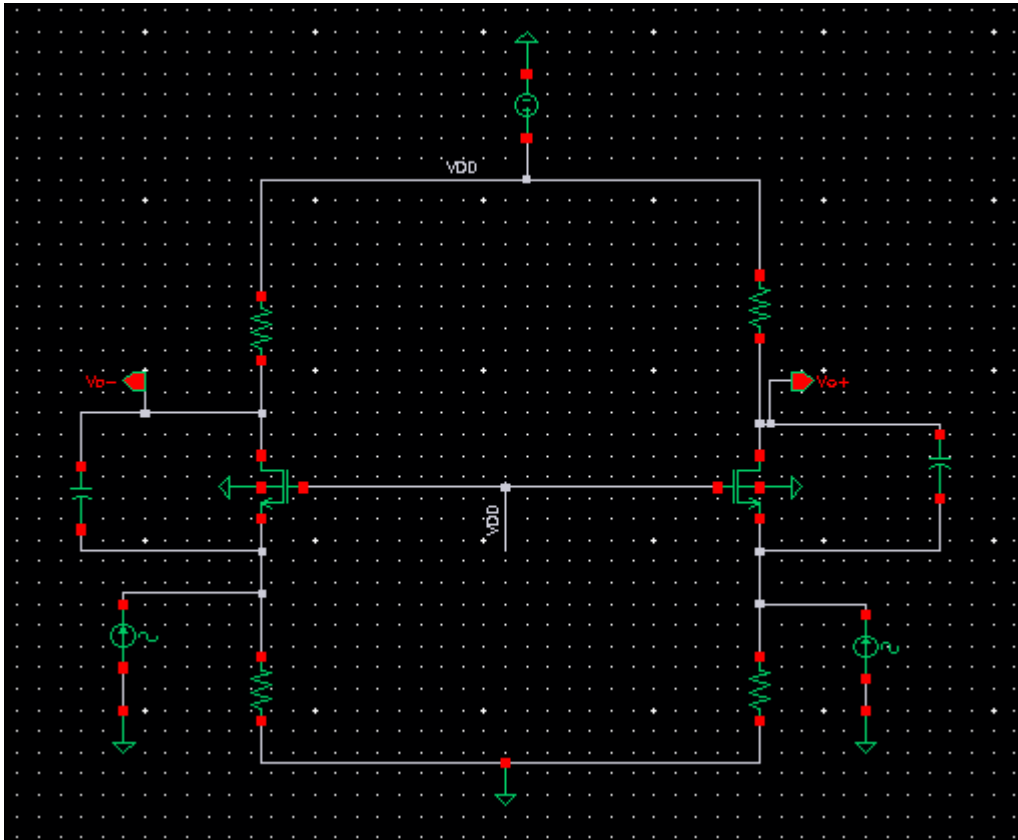
2. Exploring More Impedance Combinations for Enhanced Tunability:

Further investigation into additional combinations of impedances could lead to more flexible TI-BPF designs. By exploring a wider range of impedance options, it may be possible to create circuits with additional tunability factors, offering more control over key parameters such as center frequency, gain, and bandwidth, which would enhance the adaptability of the design to various application requirements.

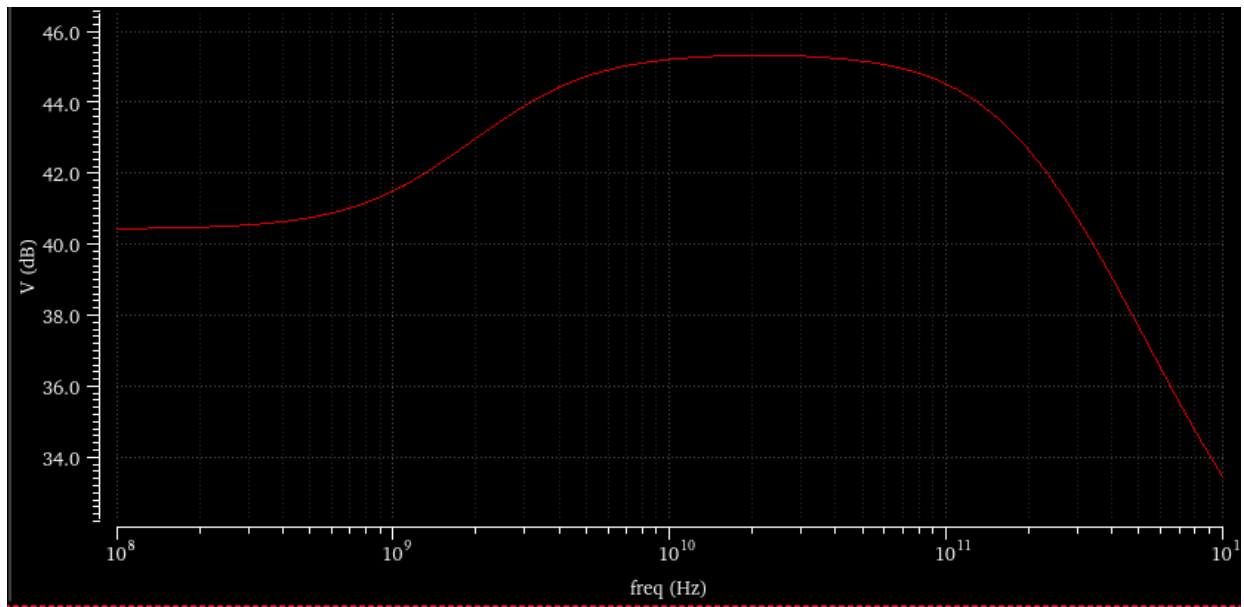
Appendix

Modification to the TIA-BPF to work as a transimpedance CTLE:

If $Z_{1, \text{ and } L}$ are modeled as resistors (active load), and Z_2 is modeled as a capacitor (see App. Fig 1), the circuit response is as the CTLE response (see App. Fig. 2), hence one circuit can work as a TIA and CTLE in high-speed data communications.



App. Fig. 1 TI-CTLE schematic



App. Fig. 2 TI-CTLE AC response

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